

0.1

VSEPR

- 2,0- Linear - 180°
 3,0- Trigonal planar - 120°
 2,1- Bent shape
 4,0- Tetrahedral - 109°28'
 3,1- Pyramidal - 107°
 2,2- Bent - 104.5°
- 5,0- Trigonal Bipyramidal
 4,1- See saw
 3,2- T-shape
 2,3- Linear - 180°
- 6,0- Octahedral - 90°
 5,1- Square pyramidal
 4,2- Square planar
- 7,0- Pentagonal bipyramidal
 6,1- Distorted octahedral

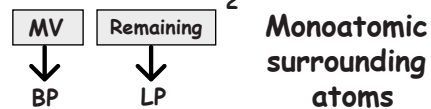
Q1

In which of the following pairs are the two species trigonal pyramidal?

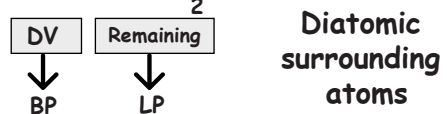
- 1) BrO_3^- and XeO_3 2) SF_4 and XeF_4
 2) SO_3^{2-} and NO_3^- 4) BF_3 and NF_3

0.2

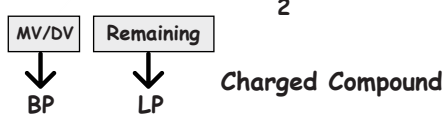
$$\text{EP} = \frac{\text{CVE} + \text{no. of MV}}{2}$$



$$\text{EP} = \frac{\text{CVE} + 0}{2}$$



$$\text{EP} = \frac{\text{CVE} + \text{negative charge} - \text{positive charge}}{2}$$



LP-LP > LP-BP > BP-BP
 (Repulsion)

Q2

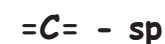
Total no. of Lone pairs in I_3^- is

- 1) 3 2) 6
 3) 9 4) 12

0.3

HYBRIDISATION

- 2- sp - Linear
 3- sp^2 - Trigonal planar
 4- sp^3 - Tetrahedral
 5- sp^3d - Trigonal pyramidal
 6- sp^3d^2 - Octahedral
 7- sp^3d^3 - Pentagonal bipyramidal



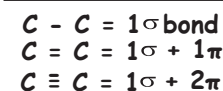
Q3

In XeF_6 , oxidation state and state of hybridisation of Xe and shape of the molecule are, respectively

- 1) +6, sp^3d^3 , distorted octahedral
 2) +4, sp^3d^2 , square planar
 3) +6, sp^3 , pyramidal
 4) +6, sp^3d^2 , square pyramidal

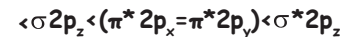
0.4

VALENCE BOND THEORY

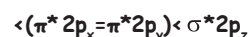
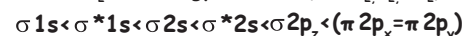


MOLECULAR ORBITAL THEORY

Up to N_2 the energy order is,



After N_2 the energy order is, (for $\text{O}_2, \text{F}_2, \text{Ne}_2$)



Q4

The no. of unpaired electrons in O_2 molecule is:

- 1) 0 2) 1
 3) 2 4) 3

0.5

BOND ORDER

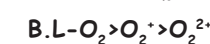
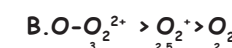
Total no. of e ⁻	2 - 1 - Diamagnetic
	3 - 0.5 - Paramagnetic
	4 - 0 - Diamagnetic
	5 - 0.5 - Paramagnetic
	6 - 1 - Diamagnetic
	7 - 0.5 - Paramagnetic
	8 - 0 - Diamagnetic
	9 - 0.5 - Paramagnetic
	10 - 1 - Paramagnetic
	11 - 1.5 - Paramagnetic
	12 - 2 - Diamagnetic
	13 - 2.5 - Paramagnetic
	14 - 3 - Diamagnetic
	15 - 2.5 - Paramagnetic
	16 - 2 - Paramagnetic
	17 - 1.5 - Paramagnetic
	18 - 1 - Diamagnetic
	19 - 0.5 - Paramagnetic
	20 - 0 - Diamagnetic

Odd number + 10 & 16 - Paramagnetic
 Even number except 10 & 16 - Diamagnetic

BOND PARAMETERS

BOND ORDER

$$\text{B.O} \propto \frac{1}{\text{B.L}} \propto \text{Stability} \propto \text{Bond strength}$$



CHEMICAL BONDING



BOND ENERGY

- 1) Hybridisation
 $\text{sp} > \text{sp}^2 > \text{sp}^3$
 2) B.E. $\propto$ B.O. (No. of Bonds)
 $\text{N}_2 > \text{O}_2 > \text{F}_2$
 3) B.E. $\propto$ E.N. difference
 $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
 4) Halogens (B.E.)
 $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$

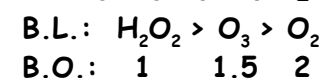
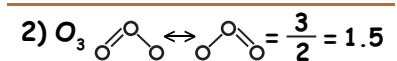
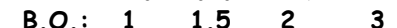
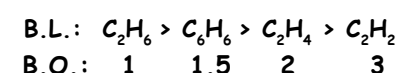
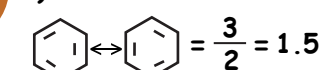
DIPOLE MOMENT (μ)

- 1) Polar molecule, $\mu \neq 0$
Irregular geometry.
 a) Different bonds
 b) Lone pairs
 c) Different surrounding atoms
 eg: SF_4 , CH_3Cl
 2) Non polar molecule, $\mu = 0$
Regular geometry
 a) Same bonds
 b) Zero lone pairs
 c) Same surrounding atoms
 eg: CH_4 , CO_2
 $\text{CH}_3\text{OH} > \text{CH}_3\text{Cl} > \text{H}_2\text{O} > \text{NH}_3 > \text{NF}_3 > (\text{BF}_3)$
 $\mu = 0$

RESONANCE

Resonance Changes B.O.

1) Benzene



BOND ANGLE

- 1) Max B.A - 180° [sp]
 2) Hybridisation
 $\text{sp} > \text{sp}^2 > \text{sp}^3$
 $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$
 $\text{sp} \quad \text{sp}^2$
 3) if LP $\uparrow$ B.A $\downarrow$
 $\text{CH}_4 > \text{NH}_3 > \text{H}_2\text{O}$ [sp³]
 0 LP 1 LP 2 LP
 4) If electronegativity of C.A $\uparrow$ B.A $\uparrow$
 $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$
 5) If electronegativity of S.A $\uparrow$ B.A $\downarrow$
 $\text{NI}_3 > \text{NBr}_3 > \text{NCl}_3 > \text{NF}_3$

BOND LENGTH

- 1) B.L $\uparrow$ with Size of atom
 $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
 2) B.L decreases with multiplicity
 $\text{C}-\text{C} > \text{C}=\text{C} > \text{C}\equiv\text{C}$

Q5

Which of the following molecules has the maximum bond enthalpy?

- 1) N_2 2) CO
 3) F_2 4) HF

Q6

Which of the following will have maximum dipole moment?

- 1) NF_3 2) NH_3
 3) CH_4 4) PCl_3

Q7

The correct order in which the O-O bond length decreases in the following is

- 1) $\text{O}_3 > \text{H}_2\text{O}_2 > \text{O}_2$ 2) $\text{O}_2 > \text{O}_3 > \text{H}_2\text{O}_2$
 3) $\text{O}_2 > \text{H}_2\text{O}_2 > \text{O}_3$ 4) $\text{H}_2\text{O}_2 > \text{O}_3 > \text{O}_2$

Q8

The correct order of bond angle is:

- 1) $\text{H}_2\text{O} > \text{NH}_3 > \text{CH}_4 > \text{CO}_2$
 2) $\text{H}_2\text{O} < \text{NH}_3 < \text{CO}_2 > \text{CH}_4$
 3) $\text{H}_2\text{O} < \text{NH}_3 > \text{CO}_2 > \text{CH}_4$
 4) $\text{CO}_2 > \text{CH}_4 > \text{NH}_3 > \text{H}_2\text{O}$

Q9

The correct order of increasing bond length of C-H, C-O, C-C & C=C is:

- 1) $\text{C}-\text{C} < \text{C}=\text{C} < \text{C}-\text{O} < \text{C}-\text{H}$
 2) $\text{C}-\text{O} < \text{C}-\text{H} < \text{C}-\text{C} < \text{C}=\text{C}$
 3) $\text{C}-\text{H} < \text{C}-\text{O} < \text{C}-\text{C} < \text{C}=\text{C}$
 4) $\text{C}-\text{H} < \text{C}=\text{C} < \text{C}-\text{O} < \text{C}-\text{C}$

Hydrogen bond- Formation of bond between hydrogen & most electronegative elements like F, O, N

Intramolecular - H - Bonding occur within one single molecule.

Intermolecular - H Bonding between two different molecules of same or different compounds.

Intramolecular H bonding increases the boiling point. eg: Intermolecular H bond in p-nitrophenol increases the boiling point. $\text{HF}(\text{HB}) > \text{HI} > \text{HBr} > \text{HCl}$. $\text{H}_2\text{O}(4\text{HB}) > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$

Fajan's Rule - Indicates Covalent character in an ionic bond.

- 1) Size of the cation- Smaller the cation higher is the polarisation, so covalent character increases
 $\text{LiCl} > \text{NaCl} > \text{KCl} > \text{RbCl} > \text{CsCl}$
 2) Size of anion- As the size of anion increases, polarisation increases, covalent character increases
 $\text{AgF} < \text{AgCl} < \text{AgBr} < \text{AgI}$
 3) Charge on the cation- As the charge increases, Covalent character also increases
 $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$